An equivalent statement, as encountered in our discussion of special relativity, is that any location in the universe can be fully specified by giving three pieces of data: where it is relative to these three spatial dimensions. In familiar language, you can specify a city address, say, by giving a street (location in the "left-right dimension"), a cross street or an avenue (location in the "back-forth dimension"), and a floor number (location in the "up-down dimension"). And from a more modern perspective, we have seen that Einstein's work encourages us to think about time as another dimension (the "future-past dimension"), giving us a total of four dimensions (three space dimensions and one time dimension). You specify events in the universe by telling where and when they occur.

This feature of the universe is so basic, so consistent, and so thoroughly pervasive that it really does seem beyond questioning. In 1919, however, a little-known Polish mathematician named Theodor Kaluza from the University of Königsberg had the temerity to challenge the obvious—he suggested that the universe might not actually have three spatial dimensions; it might have more. Sometimes silly-sounding suggestions are plain silly. Sometimes they rock the foundations of physics. Although it took quite some time to percolate, Kaluza's suggestion has revolutionized our formulation of physical law. We are still feeling the aftershocks of his astonishingly prescient insight.

Kaluza's Idea and Klein's Refinement

The suggestion that our universe might have more than three spatial dimensions may well sound fatuous, bizarre, or mystical. In reality, though, it is concrete and thoroughly plausible. To see this, it's easiest to shift our sights temporarily from the whole universe and think about a more familiar object, such as a long, thin garden hose.

Imagine that a few hundred feet of garden hose is stretched across a canyon, and you view it from, say, a quarter of a mile away, as in Figure 8.1(a). From this distance, you will easily perceive the long, unfurled, horizontal extent of the hose, but unless you have uncanny eyesight, the thickness of the hose will be difficult to discern. From your distant vantage point, you would think that if an ant were constrained to live on the hose, it would have only one dimension in which to walk: the left-right dimension along the hose's length. If